Jeffrey Liang

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Education

Brown University Providence, RI

Sc.B. in Computer Science and Economics | GPA: 3.87/4.0 *Class of 2027*

- **Relevant Coursework:** Computer Networks, Applied Cryptography, Database Management Systems, Design & Analysis of Algorithms, Machine Learning, Deep Learning, Multivariable Calculus with Theory

Experience

InstaLILY AI Sep. 2026 – Present

Software Engineer Intern (Forward Deployed Engineering)

- Shipped an AI sales assistant filtering **10K accounts** into actionable pitches for **43 reps** at a **$10B distributor**; lifted usable recommendations **55%→79%** and new-category prediction **14%→22%** via **cosine similarity**
- Moved the production assistant onto the client's own **AWS** with **zero downtime** (containers on ECS, Postgres on RDS, Temporal Cloud for job scheduling), giving the client full control of their data
- Built an AI order-intake system turning purchase-order emails into **SAP** sales orders using **Gemini**, Salesforce quote matching, and **RPA** bots

JPMorgan Chase Jun. 2026 – Aug. 2026

Equity Capital Markets Summer Analyst (Technology)

- Staffed on **2 tech IPOs** and **4 follow-on offerings**, building valuation models and trading comps analyses
- Automated tracking of **30+ comparable companies** on live deals with LLM tools (**Rogo**, internal JPM LLM)

Wlkthru Jan. 2026 – Apr. 2026

Software Engineer Intern

- Architected server-side recording for a **WebRTC** video-calling platform handling **20+ calls/week**, streaming media from a **Mediasoup** server into **FFmpeg** independent of client connections
- Eliminated video corruption under packet loss with a custom **RTP** relay that reorders out-of-order packets, keeps streams continuous across reconnects, and requests keyframes to recover
- Kept video at **720p** through camera flips (previously **480p**) by re-applying capture constraints and bitrate floors

Projects

TCP/IP Networking Stack | *Go* Feb. 2026 – Apr. 2026

- Built a user-space **IPv4** router in **Go** with longest-prefix-match forwarding and **RIP** with split horizon
- Implemented **RFC 9293 TCP** on top: handshake, the full state machine through 4-way teardown, and sliding-window flow control over 64KB circular buffers
- Ensured reliable, in-order delivery via **RFC 6298** adaptive retransmission with exponential backoff, out-of-order reassembly, and zero-window probing across goroutines

PartSelect Agent | *Python, FastAPI, React, SQLite, ChromaDB, Playwright* May 2026

- Hit **100% tool-routing accuracy** on **17 evals** with a **FastAPI** agent over **SQL + ChromaDB MCP** tools
- Scraped **7,000+ parts** via a resumable **Playwright** scraper with anti-detection, dedup, and crash recovery
- Designed a dual-layer eval harness scoring **11 categories** via deterministic routing checks and **LLM-as-judge**

Leadership

Beya: Small Business Assistance App Sep. 2024 – Apr. 2025

Technical Development Lead

- Led end-to-end development of a B2B mobile app adopted by **20+ vendors**, generating **$22K+ ARR**
- Built a serverless **AWS Lambda** payment backend processing **Stripe** webhooks across **100+ transactions**
- Architected a **React Native/Expo app** with **Firebase authentication** and real-time inventory synchronization

Skills & Interests

Languages: Python, Go, C++, JavaScript, TypeScript, Swift, SQL
Frameworks & Cloud: FastAPI, React, TensorFlow, Playwright, Temporal, AWS ECS/Lambda, GCP, ChromaDB
Interests: Tennis, Short Fiction Writing, Card Games & Probability Theory